

Geodesics of Uncharged Black Holes

Jaidev Wadhvani

1 Introduction

For a beginner in the area of computational astrophysics, geodesics of black holes are an intriguing rabbit hole to find yourself in. The best library to my knowledge regarding uncharged black holes and EMRIs is **KerrGeo.py**.

2 Benefits of using KerrGeo.py

- Orbit simulation takes place in **Kerr spacetime** [1].
- Incorporation of the **Boyer–Lindquist coordinate system** [2].
- Includes the black hole’s **angular momentum**.

3 Demerits of using KerrGeo.py

- Inability to simulate **Kerr–Newman black holes** and charged black holes as a whole.
- Interpretation of the results can be difficult.
- Primarily deals with **geodesic motion** [3].

4 Classes and Pipelines

4.1 Stable Orbit – Pipeline 1

A stable orbit is calculated using the following four parameters:

$$a, \quad p, \quad e, \quad x$$

Where:

- a = black hole spin
- p = semi-latus rectum
- e = eccentricity
- x = inclination parameter

Note: The inclination is entered in degrees and converted into x using:

$$x = \cos(i)$$

4.2 Why are we using these four parameters and not anything else?

These four parameters are enough to describe the main features of the orbit being studied. The spin a describes how fast the black hole rotates, while p and e describe the size and shape of the orbit. The inclination parameter x describes the orientation of the orbit relative to the black hole's equatorial plane.

They are therefore a convenient set of inputs for finding the constants of motion and describing a bound orbit without directly giving the energy and angular momentum as inputs.

The orbit is represented by its radial turning points:

$$r_{\text{peri}} = \frac{p}{1 + e}$$

$$r_{\text{apo}} = \frac{p}{1 - e}$$

These two values describe the closest and farthest points reached by the particle in its radial motion.

KerrGeo.py then uses the Kerr geodesic equations to determine the corresponding **energy** E , **angular momentum** L , and the **Carter constant** Q [4].

4.3 Plunging Orbit – Pipeline 2

A plunging orbit describes a particle whose trajectory takes it towards the black hole rather than remaining in a stable bound orbit.

For this pipeline, the following four parameters are used:

$$a, \quad E, \quad L, \quad Q$$

Why are we using these four parameters?

Unlike a stable orbit, where we describe the orbit using parameters such as p , e , and x , a plunging orbit is determined directly by the **constants of motion** of the particle and the spin of the black hole.

The four parameters provide the necessary information to determine the particle's trajectory:

- a describes the rotation of the black hole.
- E determines the particle's energy.
- L determines its angular momentum.
- Q accounts for the part of its motion not described by E and L , particularly motion away from the equatorial plane.[4]

Together, these quantities determine how the particle moves through Kerr spacetime. By using them, the radial and angular motion of the particle can be calculated and the trajectory can be identified as a plunging orbit.

The radial motion[5] is represented by:

$$\left(\frac{dr}{d\tau}\right)^2 = R(r)$$

Thus, the main difference between the two pipelines is simple:

- **Stable orbit:** The particle repeatedly moves between radial turning points.
- **Plunging orbit:** The particle loses its outer turning point and moves inward towards the event horizon.

5 Additional Information

1. **Kerr Black Hole:** A Kerr black hole is an uncharged, rotating black hole described by General Relativity. Its rotation causes the spacetime around the black hole to be dragged along with it.
2. **Boyer–Lindquist Coordinates:** Similar to ordinary spherical coordinates, the Boyer–Lindquist coordinate system uses four coordinates, (t, r, θ, ϕ) , where:
 - t represents time,
 - r is the radial coordinate,
 - θ is the polar angle, and
 - ϕ is the azimuthal angle.

These coordinates can be thought of as a relativistic generalization of spherical coordinates, adapted to the geometry of a rotating black hole.

3. **Geodesic Motion:** A geodesic is the straightest possible path through curved spacetime. In General Relativity, freely falling particles follow geodesics, which are determined by the geometry of spacetime.
4. **Carter Constant:** The Carter constant is a conserved quantity associated with the symmetries of Kerr spacetime. It allows the equations of geodesic motion to be separated into independent radial and angular parts, making the motion considerably easier to analyze.
5. **Radial Potential:** The radial equation of motion can be written in the form

$$\left(\frac{dr}{d\tau}\right)^2 = R(r), \tag{1}$$

where the left-hand side represents the squared radial velocity with respect to proper time τ , while $R(r)$ is known as the radial potential function. This function determines the allowed radial motion of a particle and plays an important role in studying geodesics around Kerr and Kerr–Newman black holes.